



Lab 2

1. Write a program that reads three integers and prints "all equal" if all three integers are the same, "all different" if all three integers are different, and "neither" if there is a mix of equal and different integers.

Example:

Enter 3 positive integers separated by '-': 5 - 6 - 3 all different
Enter 3 positive integers separated by '-': 5 - 5 - 5 all equal
Enter 3 positive integers separated by '-': 5 - 4 - 5 neither

2. **Create** a Java class named CustomSet to represent a set of integers. A set stores any number of distinct integers. The class will contain two private members, data and size, and two member methods, contains and union. The class will also have a constructor that takes a list of the initial set members as a parameter. The method (contains) takes an integer parameter and returns a boolean indicating whether the number is a member of the set. The method (union) takes a parameter of type CustomSet and returns the object of type CustomSet. It is important that the class includes getter and setter methods for its member variables.

Create another class named Lab2_p2 with a main function where you will create two objects of the CustomSet class. The data for both will be provided by the user. You should also create a third CustomSet which is the union of the first two, also print whether the union contains a prime number.

Example:

Please enter the members of the first set: 2, 5, 9, 7, 9
Please enter the members of the second set: -8, 3, 3, 4, 7, 1, 640
The union of the two sets is [-8,1,2,3,4,5,7,9,640]
The prime numbers that are members of the union: 2, 3, 5, 7



3. Create a class named Invoice that contains three private member variables:
- a. an integer variable named invoiceNumber.
 - b. a string variable named customerName.
 - c. an array of double variables named prices.

The class will have only one constructor that takes 3 arguments, one for each member variable. The class also contains two member methods named report and calculateInvoice. The (report) method does not accept any parameters and does not return a value. The (calculateInvoice) method takes no parameters and returns the value of type double. The (report) method will display the data of the calling object. The (calculateInvoice) method will compute and return the total cost after applying the discount based on the following table:

Discount rate %	Total cost
1	=> 1000
4	=> 5000
6	=> 8000
8	=> 10000

The class should contain getters and setters methods for each member variable. Create another class named Lab2_p3 with a main function where you will create two objects of the class Invoice and read their data from the user. Then, their data is displayed as shown in the example:



Example:

Enter account 1 data:

Invoice number: 1

Customer name: Ahmed

Number of elements: 3

Prices: 150.2, 250.36, 845

Enter account 2 data:

Invoice number: 2

Customer name: Ali

Number of elements: 4

Prices: 188.9, 365.36, 410, 145.78

The data of the first account:

Account number: 1

Name: Ahmed

Total cost: 1245.56

The total cost after discount = 1233.10

The data of the second account:

Account number: 2

Name: Ali

Total cost: 1110.04

The total cost after discount = 1098.93



4. Using the Invoice class you have defined in the previous problem, write a Java class named Lab2_p4 that contains two functions, the main function and a function called readInvoiceDetailsFromFile. The header of readInvoiceDetailsFromFile function should be: public static Invoice[] readInvoiceDetailsFromFile(String fileName).

The function should read the invoice data from the file specified by the fileName variable, store the data in an array of type Invoice (previously defined in the prior problem), and return this array. In the main function, first read the name of the file that contains the invoice data. Then, use the readInvoiceDetailsFromFile function to load this data into an array. After that, sort the array by total cost (after applying discounts) in descending order and print the results. Finally, collect new invoice data from the user and add it to the same file from which the original data was read.

The file will have the following structure:

- First line will be an integer number representing the number of invoices stored in the file.
- Each line of the next lines represents the data of one invoice comma separated.
- (Ex: the line 1,Ahmed,3,150.2,250.36,845 represent data of a banking account whose number is 1, owner name is Ahmed, and the (3 products) prices (150.2,250.36,845) which mean total cost before discount is 1245.56).



Required:

1. You are required to answer all the above questions and deliver them in your lab time.
2. A discussion will be made with you at your lab next week on what you delivered.
3. The deadline for the delivery is your lab time.

Policies:

1. You should work individually.
2. Cheating will be severely penalized, so delivering nothing is so much better than cheating.
3. No late submission is allowed.

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